

Volatility Surface & Rough Heston Pipeline

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title: “Volatility Surface & Rough Heston Pipeline” subtitle: “Progress report — Sessions A to C” author: “Akin Akandere” date: “10 September 2026”

Status

Three of six planned sessions are complete. The live implied-volatility surface is finished; the Heston characteristic function and two independent Fourier pricers are built and cross-validated; the calibrator runs and the Phase 1 baseline is recorded. Session D — the Volterra kernel and the Markovian lift — is next and has not been started.

Test suites	6
Tests passing	274 / 274
Analytical health checks	50 / 50 (16 s)
Production lines	2,268
Test lines	1,641
Commits	15
Python / numpy	3.12.3 / 2.5.2

The repository is local-only. No remote, nothing pushed.

Numerical methods

In use

Black–Scholes / Black-76 in forward measure. Everything is expressed in log-moneyness $k = \ln(K/F)$ and total variance $w = \sigma^2 \tau$, with $\tau = T - t_0$ as a time increment rather than absolute time. The normal CDF and PDF are computed locally — `scipy.stats` alone costs 5.8 s to import and this module sits on the startup path — and are verified against scipy to 2.2e-16.

Implied volatility by Newton on vega with a bisection guard. Price is monotone in σ so the bracket can never be lost, and vega is exactly the derivative required and already computed. Solved across a whole strike strip at once, which matters: the one-at-a-time version was measured at half the cost of an entire calibration objective evaluation.

Put–call parity regression for the forward. Regressing $C - P$ on K gives slope $-df$ and intercept $df \cdot F$, so both the discount factor and the forward come out of the market rather than from an assumed rate and dividend. The regression R^2 doubles as a chain-quality score. Recovered to 1.1e-13 on clean data.

Raw SVI slice fitting, scipy-free. Holding (m, σ) fixed makes the problem linear in the remaining three parameters, so the inner solve is one `lstsq` and only a two-dimensional search remains — the Zeliade quasi-explicit reduction, handled by a coarse grid plus a compass search. Recovers planted parameters to 1.4e-08.

Heston characteristic function, Albrecher branch. $g = (a-d)/(a+d)$, never the reciprocal. The broken form is kept in the codebase so the suite can measure the difference rather than assert it: at `u = 4` the max/median step across τ is 5.2 for the shipped form and 5652 for the reciprocal.

Fourier pricing, two independent routes. Lewis (no damping parameter) and Carr–Madan (damped, plus an FFT variant for a whole strike strip). They share only the characteristic function, so their agreement — 5.3e-15 — is evidence rather than restatement. Quadrature is **composite** Gauss–Legendre: one cached 64-point rule over as many panels as needed, because `leggauss` is $O(n^2)$ and cost 8.4 seconds at $n = 4000$. Truncation and panel count are derived from the characteristic function itself, since no fixed pair serves both a one-day and a one-year option.

Levenberg–Marquardt with Marquardt (diagonal) damping. Forward-difference Jacobian, bounded-logistic parameter transforms, multi-start, and a time budget. Generic over the model: a transform plus a characteristic-function factory is all it needs, so Session E's rough Heston will be calibrated by identical code.

Breeden–Litzenberger risk-neutral density. Differentiated on a dense *uniform* strike grid off the *fitted* slice — the three-point second difference loses an order of accuracy on non-uniform spacing, and real strike grids are \$1 near the money and \$5 in the wings.

Structure function $m(q, \Delta) = E|X(t+\Delta) - X(t)|^q$ with $\zeta(q) = qH$. A second, independent route to the Hurst exponent, reading the realised log-volatility path rather than the option surface. The two routes agree to 0.0029 on a common planted truth.

Monte Carlo, full-truncation Euler. Used only to validate the Fourier pricers against something that shares no machinery with them.

Davies–Harte circulant embedding for exact fractional Brownian motion in the test fixtures. An approximate generator was tried first and returned the wrong exponent, which would have looked like an estimator bug rather than a fixture bug.

Planned

- **Fractional Volterra kernel** $K(t) = t^{(\alpha-1)}/\Gamma(\alpha)$, $\alpha = H + \frac{1}{2}$, and its completely-monotone representation $K(t) = \int e^{-(xt)} \mu(dx)$ with $\mu(dx) = x^{-(\alpha)}/(\Gamma(\alpha)\Gamma(1-\alpha)) dx$. Already verified numerically to 6.7e-10.
- **Markovian lift** by a sum of exponentials, $K(t) \approx \sum w_i e^{-x_i t}$, with nodes and weights from a geometric partition of μ (Abi Jaber–El Euch). Each factor $U_i(t) = \int e^{-x_i(t-s)} dZ(s)$ obeys $dU_i = -x_i U_i dt + dZ(t)$, so the non-Markovian process becomes a finite Markovian system.
- **Lifted fractional Riccati system**, solved by RK4 — an N-dimensional set of ordinary Riccati ODEs, one per factor.

- **Hawkes self-exciting jump intensity** $\lambda(t) = \mu + \sum \alpha e^{-\beta(t-t_i)}$ over past events, in place of constant- λ Poisson.
- **Kalman and dual Kalman filtering** on an Ornstein–Uhlenbeck state space, with adaptive measurement noise across regime changes.

Literature and resources

Used so far

- Gatheral, Jaisson & Rosenbaum, *Volatility is Rough* (2014) — the structure function, $\zeta(q) \approx q^H$, and the SPX scaling regression ($\zeta(1) \approx 0.1555$, $R^2 \approx 0.93$) as the empirical anchor for $H \approx 0.1$.
- Heston (1993) for the model; Albrecher et al. (2007) for the branch-cut correction, the “little Heston trap”.
- Carr & Madan (1999) for the damped Fourier transform; Lewis (2001) for the fundamental-transform form.
- Gatheral’s SVI parameterisation and the Durrleman condition $g(k) \geq 0$ for the no-butterfly test; Gatheral–Jacquier for the necessary parameter conditions.
- Breeden & Litzenberger (1978) for the risk-neutral density.
- Lord et al. — full-truncation Euler for the Heston variance process.
- Davies & Harte — circulant embedding for exact fractional Brownian motion.
- Zeliade Systems’ quasi-explicit SVI calibration note.
- Interactive Brokers TWS API reference — tick-type semantics and pacing limits, verified directly against the installed `ibapi` rather than from memory.
- Your own lecture notes (65 slides): model specification versus parameterisation, the Kalman gain and OU state space, the structure function, fBm and fractional OU, Hawkes versus Poisson, and your handwritten derivation of the Markovian lift.

To be used

- Abi Jaber & El Euch, *Multifactor approximation of rough volatility models* — the geometric node and weight construction for the lift.
- El Euch & Rosenbaum, *The characteristic function of rough Heston models* — the fractional Riccati equation.
- Bayer, Friz & Gatheral, *Pricing under rough volatility* — rough Bergomi and the short-maturity skew.
- Shevchenko on fractional Brownian motion and Markovian lifting (not yet retrieved).
- Hawkes (1971) and Bacry–Muzy for self-exciting point processes in finance.

Health and sanity checks

Run with `python healthcheck.py`. These are distinct from the test suites: the suites assert that behaviour is correct, this reports the numbers behind those assertions, so drift stays visible even while everything still passes.

Foundations

check	measured	threshold	verdict
norm_pdf vs scipy	0	1.00e-15	PASS
norm_cdf vs scipy	2.22e-16	1.00e-15	PASS
put-call parity C-P = F-K	2.84e-14	1.00e-09	PASS
vega vs central difference	4.23e-09	1.00e-04	PASS
implied-vol round trip	8.05e-16	1.00e-06	PASS
fd_first exact on a quadratic	7.46e-14	1.00e-12	PASS
fd_second exact on a quadratic	2.08e-11	1.00e-10	PASS

Heston cf

check	measured	threshold	verdict
$\phi(0) = 1$	0	1.00e-12	PASS
$\phi(-i) = 1$ (martingale)	0	1.00e-10	PASS
$\phi(-u) = \text{conj}(\phi(u))$	0	1.00e-12	PASS
	$\phi(u)$	≤ 1	0.9999
branch continuity, shipped (max/median step) Albrecher $g=(a-d)/(a+d)$	5.242	50	PASS
branch discontinuity, reciprocal form the trap, kept to prove it is real	2825	500	PASS
trap / shipped separation	538.9 x	100 x	PASS
$\xi \rightarrow 0$ degenerates to Black-Scholes at $\xi=1e-4$; $3.5e-06$ at $\xi=1e-6$ by cancellation	5.92e-09	1.00e-08	PASS

Pricers

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check	measured	threshold	verdict
Lewis vs exact Black-76	3.55e-15	1.00e-10	PASS
Carr-Madan vs exact Black-76	1.94e-16	1.00e-10	PASS
Lewis vs Carr-Madan on Heston independent routes, 5 maturities	5.29e-15	1.00e-08	PASS
Carr-Madan insensitive to damping alpha alpha 0.75 to 3.0	6.11e-16	1.00e-08	PASS
Fourier vs Monte Carlo (sigmas) residual is Euler O(dt) bias	0.3618 sd	5 sd	PASS
simulated $E[S/F] = 1$	1.94e-06	6.12e-04	PASS
risk-neutral density mass	1	1	PASS
risk-neutral density mean = forward	1	1	PASS
density non-negativity (min/peak) tail dither scales as $1/dK^2$	-7.04e-10	-1.00e-07	PASS

Estimators

check	measured	threshold	verdict
H from ATM skew (planted 0.12) $r^2 = 1.00000$	0.12	0.12	PASS
H from structure function, worst of 4 planted $H = 0.10, 0.12, 0.30, 0.50$	0.0062	0.03	PASS
two independent H routes agree surface route vs path route	0.002891	0.05	PASS
forward from put-call parity $r^2 = 1.000000$, no rate or dividend input	1.14e-13	1.00e-06	PASS
discount factor from parity	4.44e-16	1.00e-09	PASS
SVI parameter recovery	1.43e-08	0.001	PASS
SVI slice passes Durrleman $g_{\min}$ over data = 0.2297	1	1	PASS
butterfly audit: false positives on a convex slice	0	0	PASS
butterfly audit: catches a planted dent flags the dent's neighbours	2	1	PASS
calendar audit: clean on monotone w	0	0	PASS
calendar audit: catches a reversal	3	1	PASS

Calibration

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check	measured	threshold	verdict
plant-and-recover, worst relative error 7 iters, 1.0s	3.13e-11 rel	0.01 rel	PASS
fitted RMSE on clean data	4.06e-09 vol pts	0.001 vol pts	PASS
Jacobian condition number at the solution high = parameters trade off; see identifiability	30.05	1.00e+08	PASS
spread of v0 under 0.5vp noise stable	3.775 %	10 %	PASS
spread of kappa under 0.5vp noise unidentified	34.73 %	50 %	PASS
spread of theta under 0.5vp noise stable	2.803 %	50 %	PASS
spread of xi under 0.5vp noise loose	19.13 %	50 %	PASS
spread of rho under 0.5vp noise stable	3.73 %	10 %	PASS

Session D readiness

check	measured	threshold	verdict
$K(t) = \int e^{\lambda - \lambda t} \mu(dx)$ representation $\mu(dx) = x^a - a / (\Gamma(a)\Gamma(1-a)) dx$, $a = H+1/2$	6.67e-10 rel	1.00e-08 rel	PASS

Engineering

check	measured	threshold	verdict
scipy modules on the live import path scipy.stats alone costs 5.8 s	0	0	PASS
startup to actionable error, no TWS was 9.6 s before lazy imports	0.6875 s	3 s	PASS
calibration objective evaluation 39 quotes; 470 ms before vectorising	13.69 ms	200 ms	PASS
Gauss-Legendre rules constructed 2 reuses; leggauss is $O(n^2)$	1	1	PASS
quadrature tolerance 1e-12 vs 1e-14 calibration runs at 1e-12, ~2x faster	0	1.00e-12	PASS
replay round-trip fidelity recorded vs replayed surface points	0	1.00e-12	PASS

50 of 50 checks pass (16s).

Key results so far

End-to-end surface validation. A synthetic chain is generated with a planted forward $F = S \cdot e^{(r-q)\tau}$ ($r = 4.2\%$, $q = 1.3\%$) and a planted roughness $H = 0.12$, and the pipeline is told neither. It recovers the forward to 0.0036 — 0.0006% of spot, where using spot instead would be off by 2.187 at 42 days — and $H = 0.1199 \pm 0.0009$ from the surface, 0.1178 from the realised path.

Why the coordinate change was not cosmetic. A fixed $\pm 2\%$ strike band has a width in standard deviations of $0.02/(\sigma\sqrt{\tau})$, which diverges as $\tau \rightarrow 0$: measured at 4.66σ at half a day and 0.73σ at twelve days. Vega then decays like $\exp(-d_1^2/2)$ while d_1 itself scales like $1/\sqrt{\tau}$, so the worst implied-vol uncertainty ran to **2786 vol points** at the front against 0.01 at the back — a spread of 198,906 $\times$. On a σ -normalised grid that spread is 3.6 $\times$.

Phase 1 baseline. Vanilla Heston fitted to a rough ($H = 0.12$) surface reaches 5.21 vol points RMSE, and the error is U-shaped — 7.52 at one day, 2.76 at thirty, 5.42 at a year. At one day it delivers **58%** of the market's ATM skew and goes flat below about a week. H from the market 0.120, from the fitted Heston 0.205. Fitted $\xi = 2.02$ with Feller violated by -3.58 : the model visibly contorted trying. This is the number Phase 2 must beat.

What the calibration actually determines. Under 0.5 vol-point quote noise the surface is recovered to about the noise level while the parameters are not: v_0 at 3.8% and ρ at 3.7% are stable, κ at 34.7% is not. The mechanism is proven rather than asserted — extending maturities from one year to three, past the relaxation time $1/\kappa = 0.5$ y, tightens θ from 33.0% to 5.0%. The original plan called for 5% parameter recovery under noise; that is not achievable, and asserting it would have been asserting something false.

Open flags

	severity	item
1	major	The live IBKR path has never run against a real TWS. Seven callbacks are exercised only by stubs. There is no Gateway on this machine and nothing listens on 7496/7497/4001/4002.
2	major	eSSVI is not built. <code>essvi_calendar_ok</code> measures calendar violations across independently fitted slices but nothing prevents them; a shared parameterisation would be arbitrage-free by construction.
3	moderate	<code>volatility_surface_3.py</code> is 963 lines holding six classes — IBKR callbacks, the sweeper, surface assembly, the audit, the density and the plotting. It should be split.
4	moderate	The Heston characteristic function loses precision as $\xi \rightarrow 0$ ($3.5\text{e-}06$ at $\xi = 1\text{e-}6$) through cancellation in $\kappa\theta/\xi^2$. Irrelevant at calibration-realistic ξ , but it constrains how the degeneracy limit can be tested.
5	moderate	<code>fd_second</code> drops to first-order accuracy on non-uniform grids. Documented and worked around by differentiating fitted slices on uniform grids, but a density taken straight off raw strikes would be biased.
6	minor	The test runner is hand-rolled — no pytest, no CI, no coverage measurement.
7	minor	Session D task D4, the fractional Riccati solve, is the identified schedule risk. If it slips, Session F (Hawkes) is the branch to drop.

Next

Session D: the Volterra kernel and the Markovian lift. D1 and D2 are de-risked — the completely-monotone representation is already verified to $6.7\text{e-}10$. The load-bearing test in D4 is that as $H \rightarrow \frac{1}{2}$ the lifted characteristic function converges to vanilla Heston, which is what proves the lift is a generalisation rather than a different model.